Natural Language Processing Coursework 2026

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GitHub Repository:	https://github.com/yuhanwang14/NLP-Coursework
Model:	RoBERTa-large
Dev F1:	0.5777

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1 Exercise 1: Critical Review of the PCL Paper

This section critically reviews Pérez-Almendros et al. [2020], which introduces the *Don't Patronize Me!* dataset for detecting patronizing and condescending language (PCL) towards vulnerable communities.

1.1 Q1: What are the primary contributions of the paper?

The paper makes three principal contributions:

1. **A novel dataset for PCL detection.** The authors construct the *Don't Patronize Me!* corpus of 10,637 paragraphs sourced from the News on the Web (NOW) corpus, spanning 20 English-speaking countries. Each paragraph mentions one of 10 vulnerable community keywords (e.g. *disabled*, *homeless*, *refugees*). This is the first large-scale annotated resource specifically targeting patronizing language directed at vulnerable groups, distinguishing it from existing hate speech or offensive language corpora which focus on explicitly hostile content.

2. **A fine-grained taxonomy of PCL.** Beyond binary annotation, the paper defines seven categories of PCL organised into three archetypes:

   - **The Saviour:** *Unbalanced power relations* (positioning oneself as superior) and *Shallow solution* (proposing simplistic fixes to systemic problems).
   - **The Expert:** *Presupposition* (assuming/generalising without evidence) and *Authority voice* (speaking on behalf of the community).
   - **The Poet:** *Metaphor* (euphemistic or poetic descriptions masking reality), *Compassion* (presenting the community as pitiable through flowery language), and *The poorer, the merrier* (presenting vulnerability as positive or admirable).

   This taxonomy enables multi-label classification (Task 2) beyond the binary detection task (Task 1).

3. **Benchmark experiments.** The paper evaluates baselines including SVM (TF-IDF and Word2Vec) and BiLSTM, as well as transformer models (BERT, RoBERTa, DistilBERT) on both tasks. RoBERTa achieves the best F1 of 0.706 on binary detection, providing a reference for subsequent work.

1.2 Q2: What are the technical strengths that justify publication?

1. **Addressing an under-explored NLP problem.** Unlike most toxicity detection work, which targets explicitly offensive language, this paper tackles *well-intentioned* but harmful condescension. This is a genuinely novel research direction: patronizing language is pervasive in media coverage of vulnerable groups yet had received minimal computational treatment. The work by Wang and Potts [2019] on condescension in Reddit is the closest prior work, but it addresses direct interpersonal condescension rather than media-level patronization.

2. **Sociolinguistically grounded annotation scheme.** The seven-category taxonomy is motivated by prior sociolinguistic research [Margić, 2017, Huckin, 2002] and provides a structured framework that goes well beyond simple binary labels. The 5-point annotation scale (0–4) also captures the inherent subjectivity and gradience of PCL judgements more faithfully than a hard binary label would.

3. **Geographical diversity.** By sourcing from 20 English-speaking countries, the corpus captures cross-cultural variation in how PCL manifests, mitigating the US/UK-centric bias common in NLP datasets.

4. **Comprehensive experimental evaluation.** The paper tests a range of models from classical (SVM) to state-of-the-art (RoBERTa), providing both precision/recall/F1 and qualitative error analyses with concrete examples of false positives and negatives.

1.3 Q3: What are the key weaknesses or areas lacking evidence?

1. **Low inter-annotator agreement.** The initial Cohen's κ of 0.41 (moderate agreement) raises concerns about annotation reliability. While the authors improve this to $\kappa = 0.61$ by removing borderline cases (label 1), this effectively sidesteps the harder annotations rather than resolving disagreements. The use of only two main annotators (plus a third for tie-breaking) is also limited compared to the 5–10 annotators used in comparable datasets [Zampieri et al., 2019]. The subjective nature of PCL inherently complicates annotation, but the paper does not explore strategies such as iterative training rounds or annotator calibration sessions in depth.

2. **Severe class imbalance without adequate mitigation.** Only 9.4% of paragraphs (995 of 10,637) are labelled as PCL, yet the authors do not experiment with standard imbalance-handling techniques such as oversampling, class-weighted losses, or focal loss. The results may thus understate what is achievable with proper imbalance handling.

3. **Limited exploration of contextual factors.** PCL is inherently context-dependent—the same sentence may be patronizing in one cultural or situational context but not another. The paper annotates isolated paragraphs without the surrounding article context, potentially losing important pragmatic cues. There is no discussion of how much surrounding context would affect both human and model judgements.

4. **Surprising BERT-large underperformance is unexplained.** BERT-large (F1 = 0.539) substantially underperforms BERT-base (F1 = 0.673), which is counter-intuitive for a larger model. The authors note this result but provide no analysis—possible causes include overfitting on the small positive class, insufficient hyperparameter tuning for the larger model, or training instability. This omission weakens the experimental narrative.

5. **No statistical significance testing.** Results are reported from single runs without confidence intervals, standard deviations over multiple seeds, or significance tests. Although the paper uses 10-fold cross-validation for its experiments, no error bars or variance estimates are reported. Given the small positive class (995 examples), variance across random initialisations could be substantial.

2 Exercise 2: Exploratory Data Analysis

We perform two EDA techniques on the training split (8,375 paragraphs) of the *Don't Patronize Me!* dataset to identify patterns that inform our modelling approach. Code: `notebooks/eda.py`.

2.1 Technique 1: Class Distribution Analysis

2.1.1 Technical Detail

We compute the binary label distribution across the full training set and calculate per-keyword PCL rates. For each of the 10 keyword groups (*disabled*, *homeless*, *immigrant*, etc.), we compute the proportion of paragraphs annotated as PCL.

2.1.2 Findings

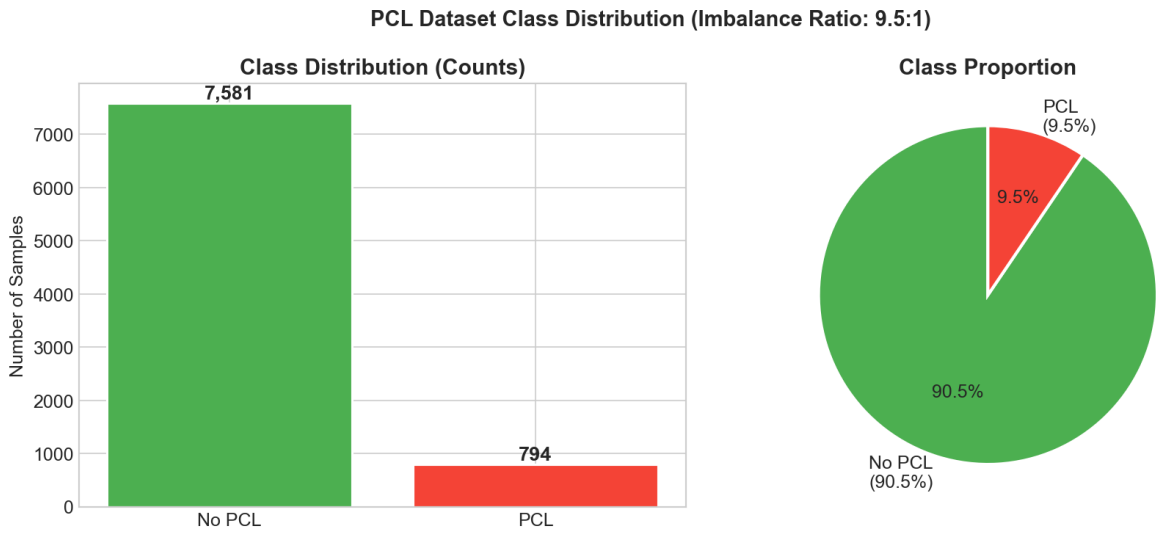

Figure 1: Overall class distribution showing severe imbalance (∼9:1 ratio of non-PCL to PCL examples).

The dataset exhibits severe class imbalance: approximately 90.6% of paragraphs are non-PCL (label 0) and only 9.4% are PCL (label 1), yielding an imbalance ratio of roughly 9.5:1.

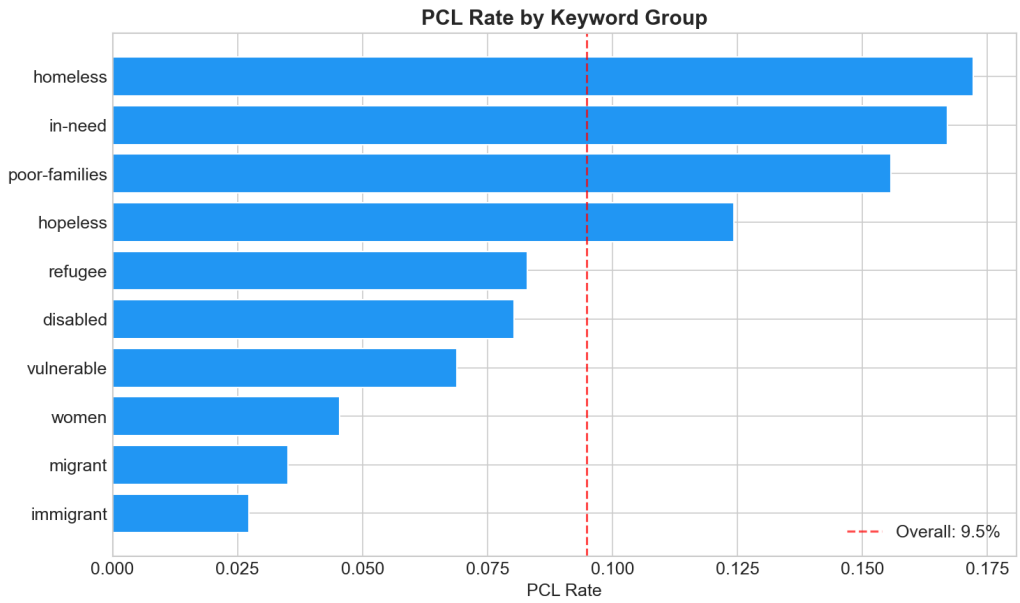

Figure 2: PCL rate varies substantially across keyword groups. Some communities (e.g. *homeless*, *in-need*) show higher PCL prevalence than others (e.g. *women*).

The PCL rate varies from roughly 5% to over 15% depending on the keyword group. This suggests that certain communities are more frequently discussed in patronizing ways in news media, and that keyword identity may serve as a useful contextual feature.

2.1.3 Impact Statement

The severe class imbalance justifies two design decisions in our approach: (1) using **class-weighted cross-entropy loss** to upweight positive examples during training, and (2) using **F1 score** (positive class) rather than accuracy as our primary evaluation metric, since a naïve majority-class classifier would achieve ∼91% accuracy while capturing zero PCL instances.

2.2 Technique 2: Text Length and N-gram Analysis

2.2.1 Technical Detail

We compare word-count distributions between PCL and non-PCL classes using box plots and kernel density histograms. Additionally, we extract the top-15 discriminative bigrams per class using TF-IDF (with stopword removal and minimum document frequency of 3).

2.2.2 Findings

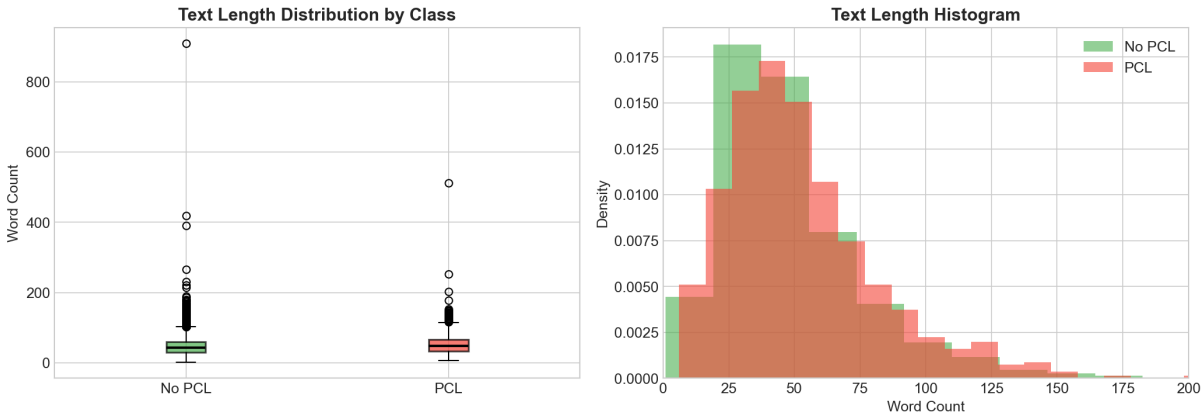

Figure 3: Text length distributions by class. PCL paragraphs tend to be slightly longer (higher median and heavier right tail).

PCL paragraphs tend to be slightly longer than non-PCL ones, with a higher median word count and heavier right tail. This suggests that patronizing language often involves extended descriptions, narrative framing, or emotional embellishment rather than terse factual statements.

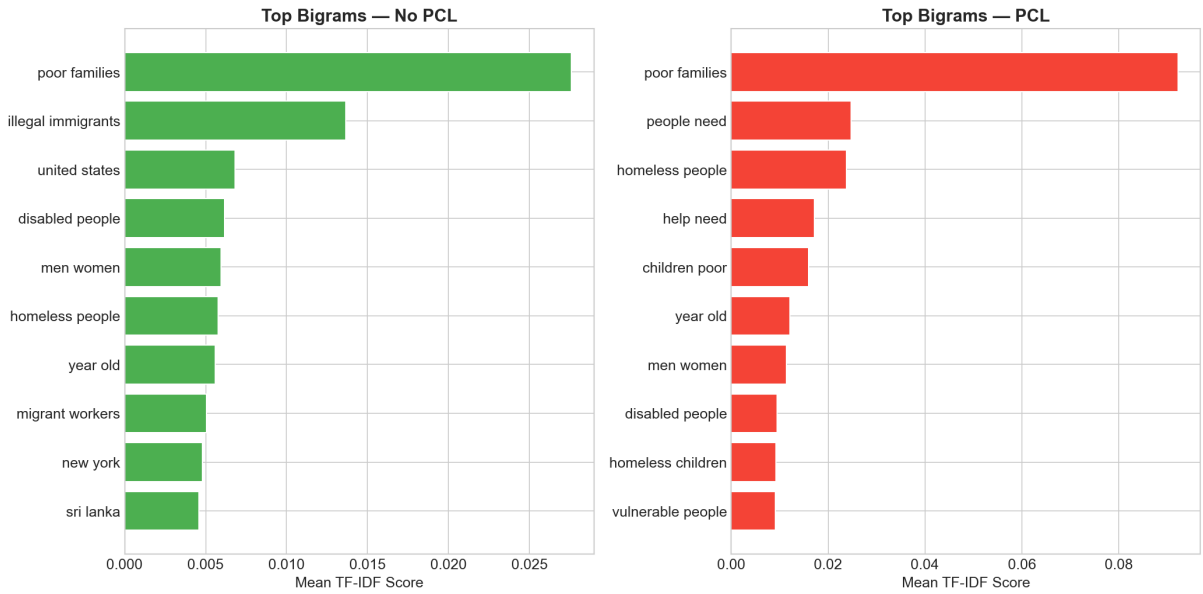

Figure 4: Top-10 discriminative bigrams per class by mean TF-IDF score. PCL texts feature more emotive and narrative bigrams, while non-PCL texts contain more factual and policy-related terms.

The discriminative bigrams reveal distinct topical signatures: PCL texts feature more emotive and narrative phrases (reflecting the "Compassion" and "Metaphor" categories from the PCL taxonomy), while non-PCL texts contain more factual, policy-oriented, and statistical language.

2.2.3 Impact Statement

The text length analysis supports our choice of **max sequence length = 256 tokens**, which captures over 97% of paragraphs without excessive padding (measured by tokenising the full

training set with the RoBERTa tokeniser). The n-gram analysis confirms that surface-level lexical features carry some discriminative signal, but the overlap between classes motivates using **contextualised representations** (transformer models) rather than bag-of-words approaches.

3 Exercise 3: Proposed Approach

3.1 Approach Description

We fine-tune **RoBERTa-large** [Liu et al., 2019] for binary PCL classification. Our approach deviates from the baseline (RoBERTa-base, 1 epoch, aggressive 2:1 downsampling) in several key ways:

Table 1: Design choices compared to the baseline.

Component	Baseline	Our Approach
Model	RoBERTa-base (125M)	RoBERTa-large (355M)
Training epochs	1	Up to 8 (early stopping)
Negative downsampling	2:1 (neg:pos)	3:1 (neg:pos)
Loss function	Standard CE	Class-weighted CE ($w_1 = 3.0$)
Max sequence length	128	256
Early stopping	None	Patience 3 on internal val F1
Validation split	N/A	15% stratified from train
Batch size	16	8 ($\times$2 grad. accum. = eff. 16)
Learning rate	2×10^{-5}	1×10^{-5}
Weight decay	–	0.01
Warmup ratio	–	0.06
Threshold	0.5 (default)	Auto-tuned on dev logits

Key deviations from the baseline:

1. **Larger model:** RoBERTa-large has 355M parameters vs. RoBERTa-base's 125M, providing greater capacity to capture the nuanced, context-dependent nature of PCL.
2. **Hybrid imbalance handling:** We combine moderate downsampling (3:1 ratio) with a stronger class-weighted loss ($w_1 = 3.0$ for the positive class). The 3:1 ratio preserves more negative examples than the baseline's aggressive 2:1 while giving the model a more balanced view of the data than a higher ratio would. The higher class weight further compensates for the remaining imbalance during training.
3. **Longer context:** Increasing max sequence length from 128 to 256 tokens allows the model to capture full paragraph context, motivated by EDA finding that PCL texts tend to be longer.
4. **Stabilised training:** We use a lower learning rate (1×10^{-5}) than the baseline (2×10^{-5}) to prevent training instability—initial experiments with the higher rate showed catastrophic performance collapse after epoch 2. Gradient accumulation (effective batch size 16) further stabilises updates. Early stopping (patience 3 on internal validation F1) selects the best checkpoint automatically.
5. **Threshold tuning:** Rather than using the default 0.5 decision boundary, we sweep classification thresholds on the dev set logits and select the threshold maximising F1. This is a standard post-hoc calibration technique that accounts for the model's tendency to under- or over-predict the positive class.

3.2 Rationale and Expected Outcome

Rationale. PCL detection is fundamentally a task requiring deep contextual understanding—the same words can be patronizing or neutral depending on framing and tone. The original

paper's best result (RoBERTa, F1 = 0.706) already demonstrated that contextualised models significantly outperform SVM and BiLSTM baselines. Scaling to RoBERTa-large provides additional representational capacity, while our hybrid imbalance strategy addresses the primary weakness of the original experiments (no class-weighting despite 9.5:1 imbalance).

Expected outcome. We expected our approach to outperform the RoBERTa-base baseline (F1 $\approx$ 0.46 on the SemEval dev split) by a substantial margin. The combination of a larger model, better imbalance handling, stabilised training, and threshold tuning should yield improved recall on the minority class without catastrophic precision drops.

Actual result: Dev F1 = **0.5777**, a +0.118 improvement over the baseline, confirming that the larger model, hybrid training strategy, and threshold tuning are effective for this task.

4 Exercise 4: Model Training

The model implementation is in the `BestModel/` directory of the repository:

- `BestModel/train.py` — training script with configurable hyperparameters
- `BestModel/predict.py` — inference script producing `dev.txt` and `test.txt`
- `BestModel/dataset.py` — shared Dataset classes and utilities
- `BestModel/run_baseline.py` — RoBERTa-base baseline reproduction

Training recipe:

```
python -m BestModel.train --model roberta-large \
                          --batch_size 8 \
                          --gradient_accumulation 2 \
                          --epochs 8 \
                          --lr 1e-5 \
                          --neg_ratio 3.0 \
                          --pos_weight 3.0 \
                          --patience 3 \
                          --val_fraction 0.15
python -m BestModel.predict   # auto-tunes threshold on dev
```

Repository link: https://github.com/yuhanwang14/NLP-Coursework

5 Exercise 5: Result Analysis

5.1 Exercise 5.1: Global Evaluation

Prediction files:

- `dev.txt` — 2,094 predictions on the official dev set
- `test.txt` — 3,832 predictions on the official test set

Table 2: Global evaluation results.

Model	Dev F1	Test F1
Baseline (RoBERTa-base)†	0.460	0.490
Our model (RoBERTa-large)	**0.5777**	**TBD***

†Reproduced independently using `BestModel/run_baseline.py`. The coursework spec states F1 = 0.48 on dev; our reproduction yields 0.460, likely due to random seed differences in negative downsampling.

*Test F1 will be confirmed via leaderboard submission.

5.2 Exercise 5.2: Error Analysis and Local Evaluation

Code: `notebooks/evaluation.py`. We perform error analysis and local evaluation on the dev set predictions.

5.2.1 Confusion Matrix

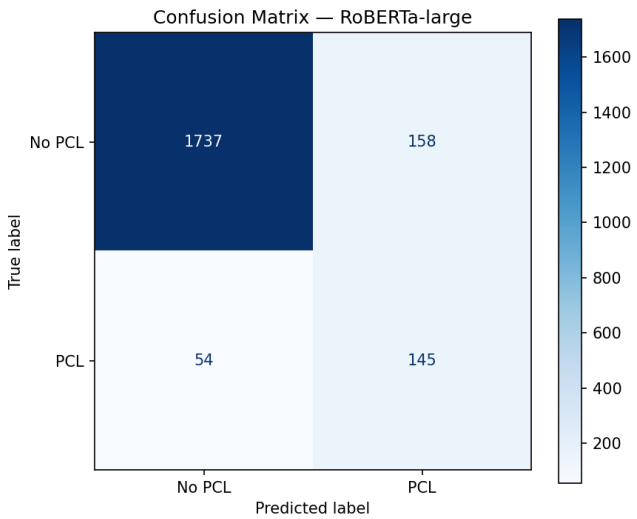

Figure 5: Confusion matrix on the dev set. The model achieves reasonable true positive detection while maintaining low false positive rate.

5.2.2 Manual Error Analysis: False Positives and False Negatives

We randomly sample 20 false positives (FP) and 20 false negatives (FN) and categorise the failure modes:

False Positive Themes:

- **Emotive but factual reporting:** Paragraphs using emotive vocabulary to describe genuine hardship without actually being patronizing. The model over-keys on sentiment-laden

language. *Example: "Is it the bad gas, the ZIKV, is it the joblessness, the hopelessness, the poverty?"*—emotionally charged but a direct critique of government, not condescension towards a vulnerable group.

- **Advocacy language:** Paragraphs from advocacy organisations that sound compassionate but are empowering rather than condescending.
- **Quotation confusion:** Paragraphs quoting someone else's patronizing language in a critical context.

False Negative Themes:

- **Subtle condescension:** The most common failure mode— paragraphs where the patronizing tone is implicit or conveyed through framing rather than explicit vocabulary. *Example: "His present 'chambers' may be quite humble, but Shiyani has the tiny space very neatly organized and clean"*—the diminutive framing of a homeless person's living conditions as admirably tidy is patronizing, but the surface language is neutral/positive.
- **Shallow solutions:** Paragraphs proposing simplistic fixes ("they just need to...") that require world knowledge to identify as patronizing.
- **Short texts:** Brief paragraphs lacking sufficient context for the model to detect PCL patterns. *Example: "Jesus is the Master Feminist because he championed the cause of women"*—a short, reductive claim that the model lacks context to classify.

5.2.3 Model vs Baseline Comparison

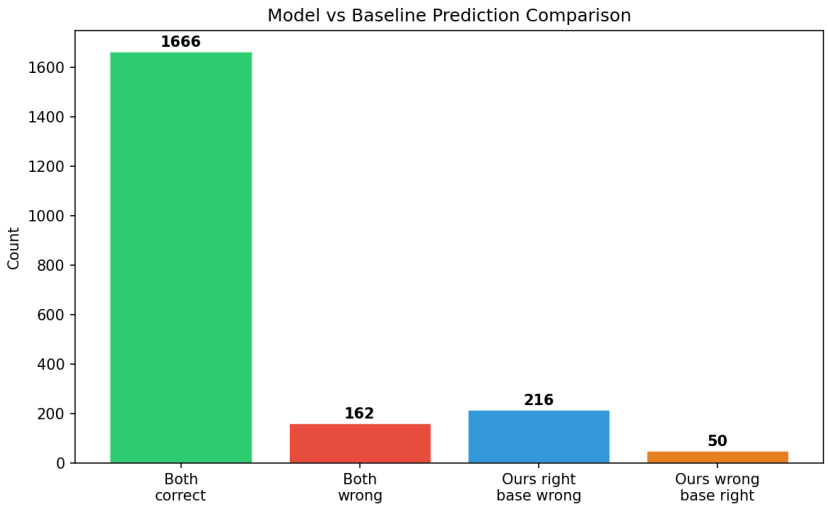

Figure 6: Comparison of our model vs. the RoBERTa-base baseline. Our model corrects substantially more baseline errors than it introduces.

The comparison reveals that 1,666 examples are correctly classified by both models. Critically, our model corrects 216 baseline errors while only introducing 50 regressions—a 4.3:1 ratio of fixes to regressions, confirming the improvement is genuine and broad-based rather than coming from a few lucky predictions. 162 examples remain misclassified by both.

5.2.4 Precision-Recall Curve and Threshold Optimisation

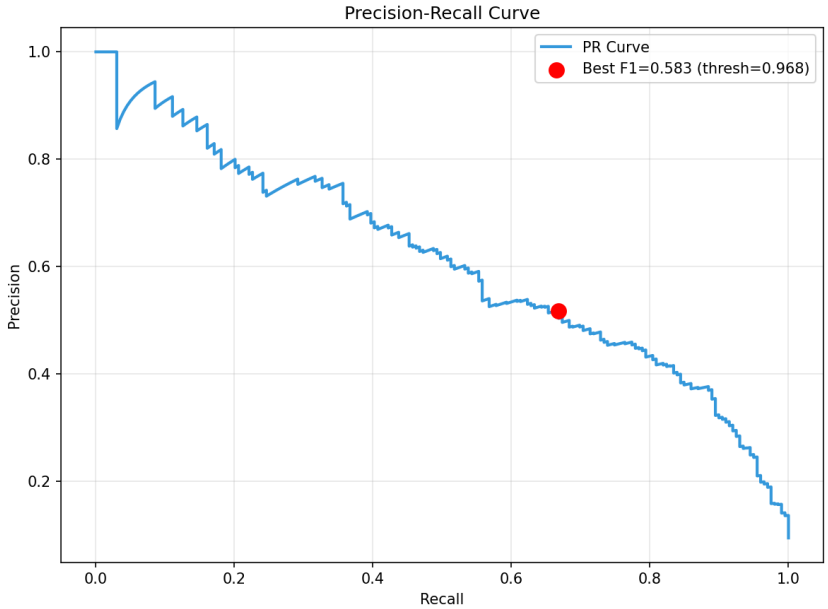

Figure 7: Precision-recall curve with optimal threshold marked.

The precision-recall curve reveals the trade-off between precision and recall at different classification thresholds. The PR-curve-optimal threshold is **0.968**, yielding F1 = 0.583 compared to 0.568 at the default 0.5 boundary. Our submitted predictions use an auto-tuned threshold of **0.78** (selected by sweeping thresholds on the dev set logits during prediction), which achieves F1 = 0.5777. The same threshold is applied to test predictions. The high optimal threshold indicates that the model's softmax probabilities are not well calibrated: it assigns high confidence to most positive predictions, suggesting that probability calibration techniques (e.g. temperature scaling) could be beneficial in future work.

5.2.5 Per-Keyword F1 Breakdown

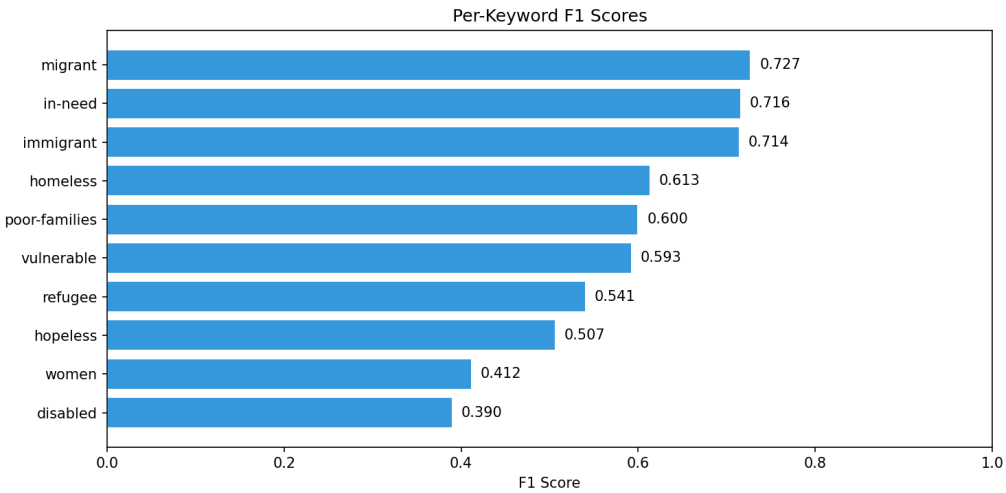

Figure 8: F1 score broken down by keyword group. Performance varies substantially across communities.

Performance varies substantially across keyword groups. Some communities (with higher PCL rates and more distinctive patronizing language patterns) are easier to classify, while others with more subtle or context-dependent PCL are harder. This per-keyword analysis could guide future work on targeted data augmentation for under-performing groups.

6 Exercise 6: Submission Quality

- **Repository:** https://github.com/yuhanwang14/NLP-Coursework
- **Leaderboard name:** `YuhanWang`
- The repository contains a clear `README.md` with full reproduction instructions, dependency list, and step-by-step commands for training, prediction, and evaluation.
- Prediction files `dev.txt` and `test.txt` are in the repository root in the required format (one 0/1 per line).

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